Multi-Agent Reinforcement Learning for Traffic Flow Management of Autonomous Vehicles

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Introduction & Problem Statement

- Traditional traffic systems fail to handle rapid urban traffic growth
- Fixed-cycle signals ignore real-time road conditions
- Results in congestion, delays, energy waste, and accident risks
- Autonomous Vehicles need intelligent coordination
- AI-driven Intelligent Transportation Systems are required

Objectives & Contributions

- Optimize autonomous vehicle traffic flow
- Apply Multi-Agent Reinforcement Learning (IA2C, MA2C)
- Formulate signal control as a Markov Decision Process
- Introduce smart rerouting for congestion balancing
- Validated using SUMO simulation

Related Work & Baselines

- Fixed-cycle signals are inefficient for dynamic traffic
- Single-agent RL lacks coordination capability
- Existing MARL focuses only on intersections
- Proposed work integrates signal control with global rerouting

Methodology Overview

- Each intersection acts as a learning agent
- IA2C: Independent agent learning
- MA2C: Neighbor-aware cooperative learning
- SUMO used for realistic traffic simulation
- Dijkstra-based dynamic rerouting strategy

Environment Setup

- 7-intersection urban road network
- Traffic lights with dynamic phase control
- Lane detectors and rerouting devices
- 7,200 vehicles per episode
- State: queues, density, waiting time

Core Technical Contributions

- MA2C enables inter-agent communication
- Spatial reward discounting for coordination
- IA2C used as baseline comparison
- Smart rerouting prevents upstream congestion

Training Details

- Artificial Recurrent Neural Network (A-RNN)
- CNN + LSTM architecture
- Centralized training, decentralized execution
- Gamma = 0.99, Learning Rate = 0.01
- 200 training episodes

Performance Results

- Fixed Signal Travel Time: ~13,076 s
- IA2C: 10,152 s (17% improvement)
- MA2C + Rerouting: 8,036 s (38% improvement)
- MA2C achieves faster convergence

Sustainability Impact

- Reduced congestion and idle time
- Improved fuel efficiency
- Lower vehicular emissions
- Enhanced energy efficiency

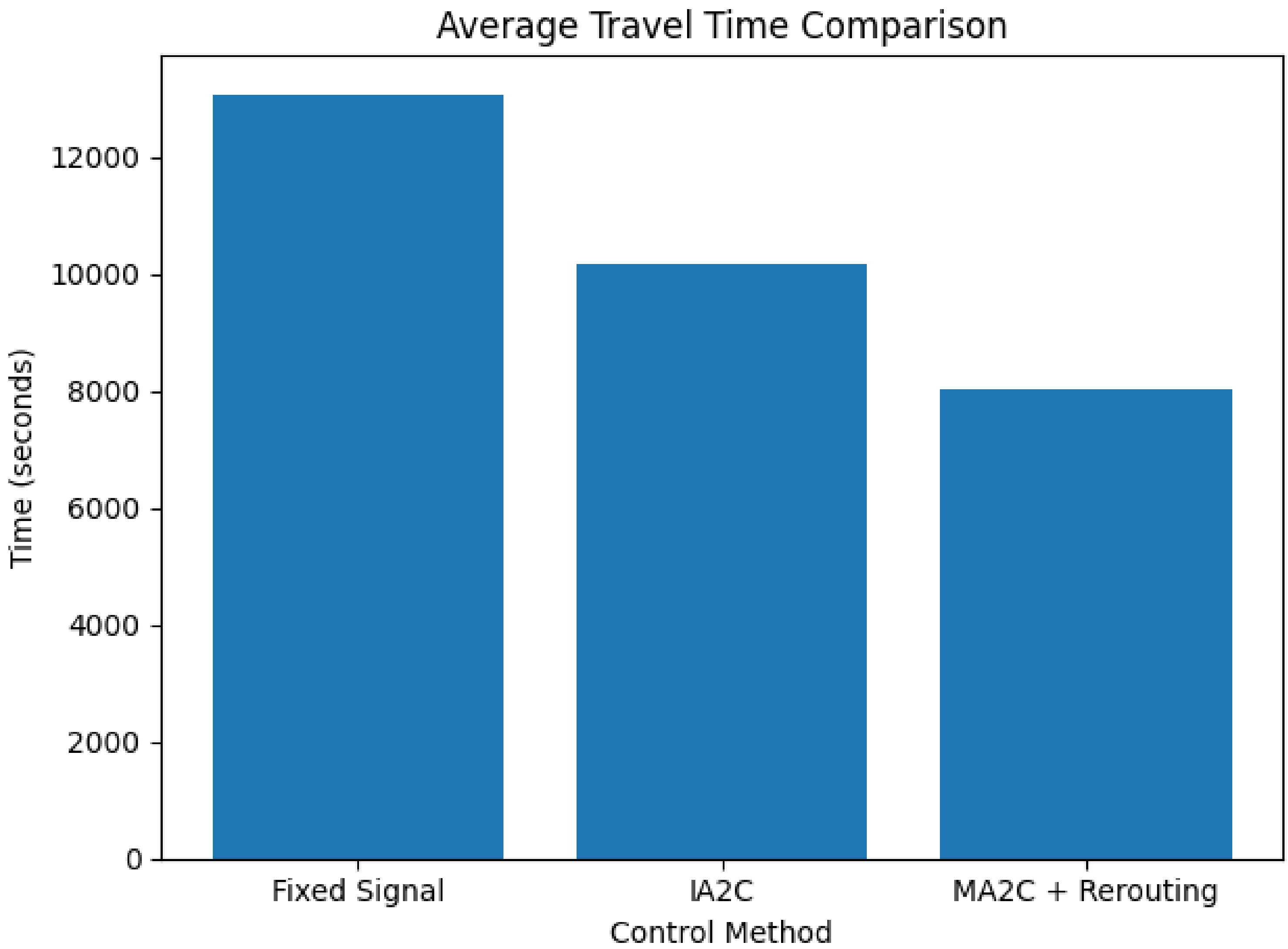
Discussion

- MA2C benefits from shared local observations
- Rerouting balances network-wide traffic load
- LSTM handles non-Markovian environments
- Entropy regularization improves exploration

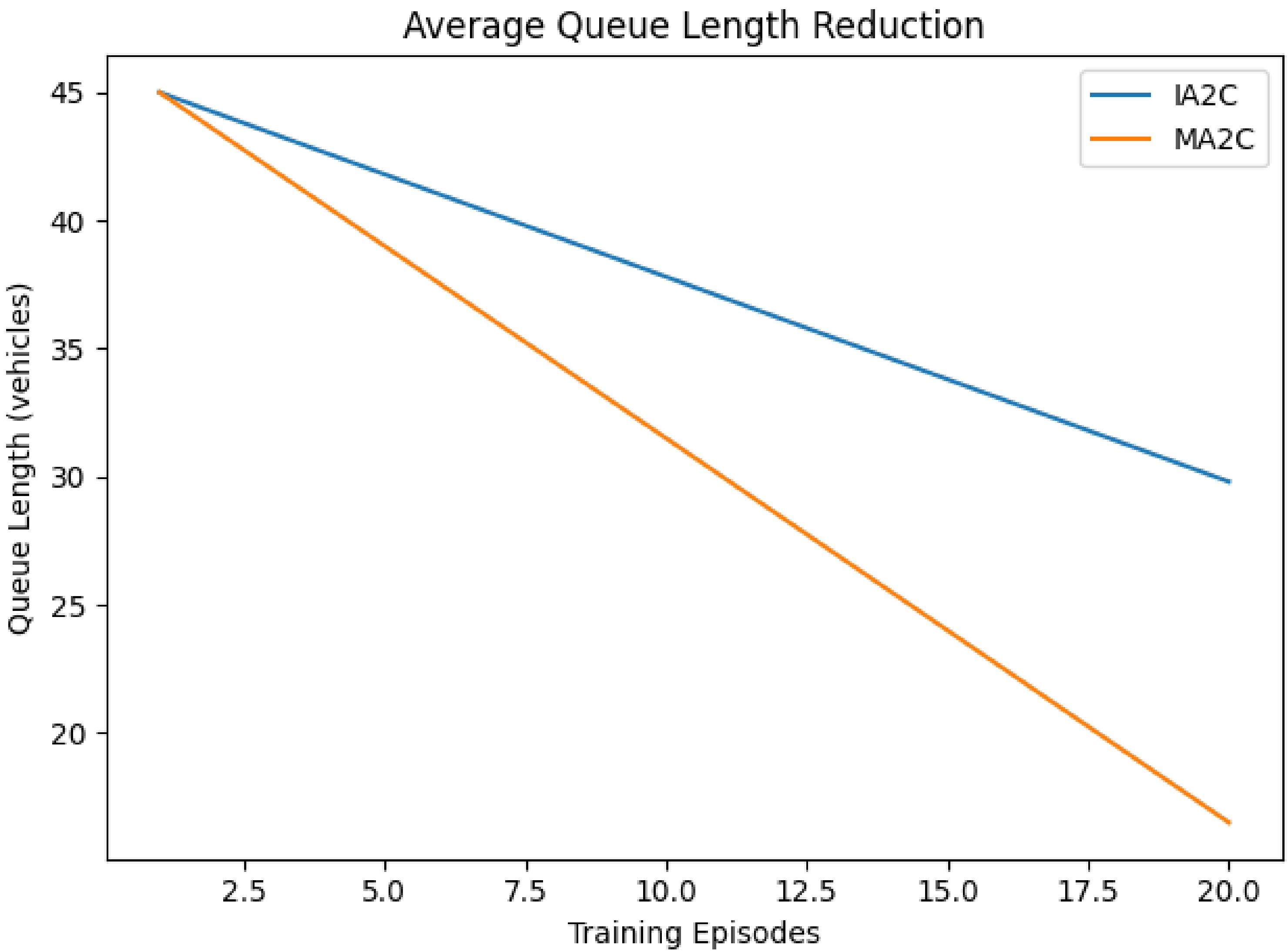
Conclusion & Future Work

- MA2C with rerouting outperforms traditional systems
- Efficient handling of large state-action spaces
- Future: larger networks and real traffic data
- Integration with V2X communication

Travel Time Comparison



Queue Length Reduction



Training Convergence